

6.6 Prototype Development Cycle

پروٹوٹائپ ایک ہی بار میں نہیں بنتا؛ اسے بنایا، آزمایا، بہتر کیا اور دوبارہ آزمایا جاتا ہے۔

CS XII

Chapter 6: Entrepreneurship in the Digital Age

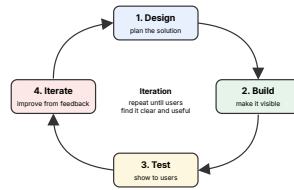
Lecture 07 of 18

★ Golden

6.6 Prototype Development Cycle ★ Golden

LEARN IT

Prototype development is **not a one-time activity**. A prototype is **created, tested, improved, and tested again**. This repeating process is called **iteration**. The main stages are **Design, Build, Test, and Iterate**.



The prototype development cycle: Design → Build → Test → Iterate, then back to Design.

Stage	What happens
1. Design	Understand the problem and plan the solution. Decide what the prototype should show (e.g. screen flow, features).
2. Build	Create the first version (physical or digital). The aim is not beauty; the aim is to make the idea visible.
3. Test	Show the prototype to users. Observe what they understand and what they find difficult.
4. Iterate	Improve the prototype based on user feedback. Remove unnecessary parts and add missing features.

WORKED EXAMPLE

Worked example: two rounds of the cycle. A team in Thatta wants a WhatsApp catalogue for a local Ajrak shop.

Round 1

- **Design:** The problem is that customers do not know which designs and prices are available. Plan: one page per design with photo, price and an "Order" message.
- **Build:** They make a paper booklet with 6 hand-drawn "catalogue pages".
- **Test:** They show it to 5 customers. Three ask, "Is this single or double-sided Ajrak? What size?"
- **Iterate:** They add size and type to every page.

Round 2

- **Design:** Add the size and type fields, and a delivery note.
- **Build:** They make the catalogue in WhatsApp Business with real photos.
- **Test:** 5 new customers browse it. Now nobody asks about size; one says the delivery charge is unclear.
- **Iterate:** They add "Delivery in Thatta: Rs 100" to each item.

Each round produces a clearer version, because each change comes from real feedback.

REMEMBER IT

Mnemonic: "Do Big Things Iteratively" = Design, Build, Test, Iterate. And remember the arrow at the end goes **back to Design**: it is a **cycle**, not a line.

COMMON MISTAKE

Mistake 1: Drawing the stages as a straight line that ends at "Iterate". **Fix:** Always draw a loop with an arrow from Iterate back to Design.

Mistake 2: Making the Build stage beautiful. **Fix:** The source says "The aim is not beauty; the aim is to make the idea visible."

اردو میں سمجھیں

پروٹوٹائپ ایک ہی بار میں نہیں بنتا؛ اسے بنایا، آزمایا، بہتر کیا اور دوبارہ آزمایا جاتا ہے۔ اس بار بار دہرانے کے عمل کو **ایٹریشن** (Iteration) کہتے ہیں۔ چار مراحل ہیں: ڈیزائن (مسئلہ سمجھ کر حل کی منصوبہ بندی)، بلڈ (پہلا نمونہ بنانا)، ٹیسٹ (صارفین کو دکھانا) اور ایٹریت (رائے کی بنیاد پر بہتری)۔

EXAM TIP

A favourite long question: "Explain the prototype development cycle with a diagram." Draw the four boxes in a **circle** with arrows, write one or two lines for each stage using the source's words, define **iteration**, and finish with a short example (e.g. the ClassBuddy activity in 6.7). Marks are often lost for drawing a straight line instead of a cycle.

CHECK YOURSELF

1. What is iteration?
2. At which stage do you observe what users find difficult?
3. What is the aim of the Build stage?

Answers: 1. The repeating process of creating, testing, improving and testing a prototype again. 2. The Test stage. 3. To make the idea visible (not to make it beautiful).

Next lecture: 6.7 Class Activity: Create a Prototype